Gender Pay Gap

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Set-up your environment

Title Page

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Part 1 - Identify a Social Problem

1.1 Describe the Social Problem

The gender pay gap (GPG) remains an important social and economic issue in the European Union (EU). Although certain progress has been made toward improved gender equality, women still earn less than men on average in many European countries (Eurostat, 2024). Although the gender pay gap differs between countries and regions, differences in earnings between men and women still continue to be a problem across Europe.

Several factors lead to the gender pay gap. This includes things such as differences in working hours, segregation by occupation, career interruptions related to pregnancy and caregiving responsibilities, and unequal representation in higher-paying positions (Blau & Kahn, 2017). Recent research also reveals the fact that structural inequalities in labour markets and workplaces continue to influence earnings differences between men and women (International Labour Organization [ILO], 2024).

Understanding the gender pay gap is very important, since it affects women' s job opportunities, financial independence and also long-term economic security. Lower earnings throughout a woman's career can also lead to lower pension income and an increased risk of poverty later in life (European Commission, 2024). Also, wage inequality can have a negative impact on the economy because women may not always be able to fully use their skills and reach their full potential.

This study examines the gender pay gap across the EU by grouping Northern, Western, Southern, and Eastern European countries, and then comparing individual regions. Using wage, employment, and income data collected over multiple years, our research aims to identify regional patterns and trends in gender-based wage inequality.

Part 2 - Data Sourcing

2.1 Load in the data

2.2 A short summary of the datasets we used

The gdp_data dataset summary:

The dataset "Gender pay gap in unadjusted form" comes from Eurostat. It measures the unadjusted gender pay gap, which is the difference in average gross hourly earnings between men and women. The dataset was last updated 3 months ago and contains 584 data cells.

- freq: Only one value is used, which is A (annual), so there is only one unique value.
- unit: Only PC (percentage) is used, so there is one unique value.
- nace_r2: Only the category [B-S_X_O] is included, so there is one unique value.
- geo: There are 39 geo codes instead of 41, meaning some geographic codes are missing from the extracted data. The missing codes seem to be: "LI" (Liechtenstein) and "EA21".
- TIME_PERIOD: The data covers 2007 to 2024.
- values: These represent the percentage difference in hourly wages between men and women.

The salary_data dataset summary:

The dataset "Average full time adjusted salary per employee" comes from Eurostat. This dataset shows the average salary per employee (full-time adjusted), allowing fair comparisons between countries regardless of differences in part-time work. The dataset was last updated 7 months ago and contains 1,560 data cells.

- freq: Only one value is used, A (annual), so there is one unique value.
- unit: Two units are used, EUR (euro) and NAC (national currency), so there are two unique values.
- geo: Includes 26 European countries and 2 additional aggregates, resulting in 28 unique geo entries. - TIME_PERIOD: The data covers the years from 1995 to 2024.
- values: Salaries are expressed in EUR and national currency (NAC).

```
##        freq             unit            nace_r2              geo
##  Length   :584   Length   :584   Length   :584   Length   :584
##  N.unique :  1   N.unique :  1   N.unique :  1   N.unique : 39
##  N.blank  :  0   N.blank  :  0   N.blank  :  0   N.blank  :  0
##  Min.nchar:  1   Min.nchar:  2   Min.nchar:  7   Min.nchar:  2
##  Max.nchar:  1   Max.nchar:  2   Max.nchar:  7   Max.nchar:  9
##
##   TIME_PERIOD             values
##  Min.   :2007-01-01   Min.   :-1.30
##  1st Qu.:2011-01-01   1st Qu.: 9.70
##  Median :2016-01-01   Median :14.45
##  Mean   :2015-08-06   Mean   :13.87
##  3rd Qu.:2020-01-01   3rd Qu.:17.60
##  Max.   :2024-01-01   Max.   :30.90
```

```
##        freq              unit               geo          TIME_PERIOD
##  Length   :1560   Length   :1560   Length   :1560   Min.   :1995-01-01
##  N.unique :   1   N.unique :   2   N.unique :  28   1st Qu.:2004-01-01
##  N.blank  :   0   N.blank  :   0   N.blank  :   0   Median :2011-01-01
```

```
##  Min.nchar:   1   Min.nchar:   3   Min.nchar:   2   Mean   :2010-06-16
##  Max.nchar:   1   Max.nchar:   3   Max.nchar:   9   3rd Qu.:2018-01-01
##                                                    Max.   :2024-01-01
##      values
##  Min.   :    964
##  1st Qu.:  14585
##  Median :  25700
##  Mean   :  97584
##  3rd Qu.:  38114
##  Max.   :7297665
```

2.3 Description of the type of variables included

Variables included in salary_data:

- freq: This variable shows how often the data is collected over time. In this dataset, it is recorded annually (A).
- unit: The unit of measure variable shows the measurement scale used for the data values. The unit of measures used in this dataset are EUR (euro) and NAC (National currency).
- geo: The variable geopolitical entity identifies 28 territorial units. It includes 26 individual European countries, represented by ISO 3166-1 alpha-2 codes, as well as two supranational aggregates.
- TIME_PERIOD: The time variable shows the time period covered by the dataset, which spans from 1995 to 2024. It is recorded in annual intervals using the YYYY format.
- values: the variable value is the annual salary in both euro (EUR) and the national currency (NAC).

Variables included in gpg_data:

- freq: This variable shows how often the data is collected over time. In this dataset, it is recorded annually (A).
- unit: shows the measurement scale used for the data values. The unit of measures used in this dataset is percentage (PC).
- nace_r2: The variable nace_r2 identifies the economic sector according to the NACE Rev. 2 classification. The category B-S excluding O includes industry, construction, and service activities, while excluding public administration, defence, and compulsory social security.
- geo: The variable geopolitical entity 41 territorial units. It includes 37 individual European countries, represented by ISO 3166-1 alpha-2 codes, as well as four supranational aggregates.
- TIME_PERIOD: The time variable shows the time period covered by the dataset, which spans from 2007 to 2024. It is recorded in annual intervals using the YYYY format.
- values: The variable value represents the unadjusted Gender Pay Gap (GPG), which is the percentage difference in average gross hourly earnings between male and female employees, relative to male earnings.

Part 3 - Quantifying

3.1 Data cleaning

In this section, we clean the Gender Pay Gap and salary datasets to prepare them for analysis. First, we check for missing values and remove incomplete In this section, we clean the Gender Pay Gap and salary datasets using several data cleaning steps to improve data quality and comparability before conducting the analysis. Firstly, we remove missing values because they may affect the reliability of the results.However, this also reduces the sample size, potentially affecting the accuracy of the results.

Secondly, we keep only the variables needed for the analysis: country, year, and values. For the salary dataset part, we retain only observations reported in euros (EUR). This ensures that all salary values are expressed in a common currency and can be compared consistently across countries.

Thirdly,we removed aggregate regions such as EA19, EA20, and EU27_2020 because our analysis focuses on individual European countries. This helps avoid double counting and makes regional comparisons clearer. However, it also means that our results cannot be directly compared to EU-wide averages

Finally, we renamed the value variables and merged the Gender Pay Gap and salary datasets by country and year. This allowed us to calculate GPG Severity, although only observations available in both datasets could be retained in the final dataset.

3.2 Generation of Necessary Variables

Variable 1 – `region`

- This is a categorical variable, which assigns each EU country to one of the four European regions: Northern, Western, Southern or Eastern Europe.
- The classification we used follows the United Nations 'geo-scheme', and is aimed to uncover and show broader geographical patterns in the gender pay gap.
- By grouping EU countries into regions, we can compare the average pay gaps and salary levels across blocks which are both culturally and economically similar, rather than simply analysing individual countries in isolation.

Variable 2 – `gpg_severity`

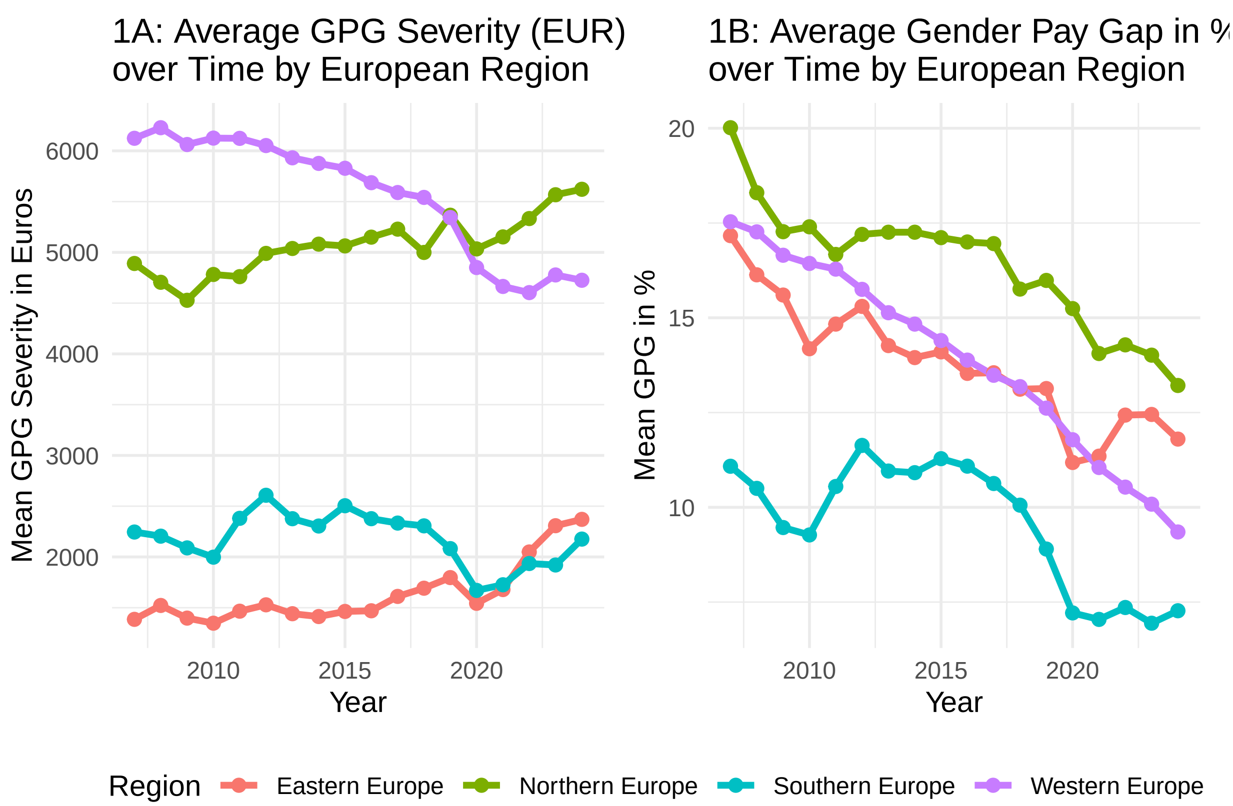
- This is a continuous variable, which translates the relative gender pay gap (which expressed as a percentage in the data-set) into an absolute difference in annual earning in euros between women and men.
- It is calculated with the formula `gpg_severity = (gpg / 100) * salary`, where `gpg` is the unadjusted gender pay gap, expressed as a percentage, and `salary` is the average full-time adjusted salary per employee, expressed in euros.
- Thus, the resulting value gives us the approximate amount in euros, by which the average annual earnings of women are lower than that of men.
- The monetary severity this variable introduces to us allows us to assess the economic impact on women across different European regions and associated years, which we would be unable to assess when only looking at the percentage gap indicating relative inequality.

Sub-variable – `region_summary`

- This is not a variable on its own, it is rather a set of sub-variables relating to variable 2, which will make all of our graphical analysis possible from here on.
- For each region and year, this summary set computes the mean gender pay gap (`mean_gpg`), the mean salary (`mean_salary`), the mean gender pay gap severity (`mean_gpg_severity`), and the number of countries included (`n_countries`). These regional aggregates are thus used for the creation of the temporal and spatial variation visualisations further below in this report.

3.3 Visualize temporal variation

The graph for Visualisation of Temporal Variation consists of two line plots. The evolution of the Gender Pay Gap over time (x-axis) for the four European regions (coloured lines) is shown in both graphs.



Plot 1A: Average Gender Pay Gap Severity in Euros over Time

- This plot shows the average/mean Gender Pay Gap Severity in Euros per Region per Year. GPG Severity, as we already discussed, shows the absolute gap in earnings between women relative to men, calculated with our formula: `gpg_severity = (gpg / 100) * salary`. When we express this gap in monetary terms, as can be seen in this plot, the real economic disadvantage women face is highlighted. A higher line therefore indicates a larger absolute gap in earnings. Further, this plot also allows us to see whether the impact of the pay gap has been growing, decreasing or remained constant over time across European regions.

Plot 1B: Average Gender Pay Gap in % over Time

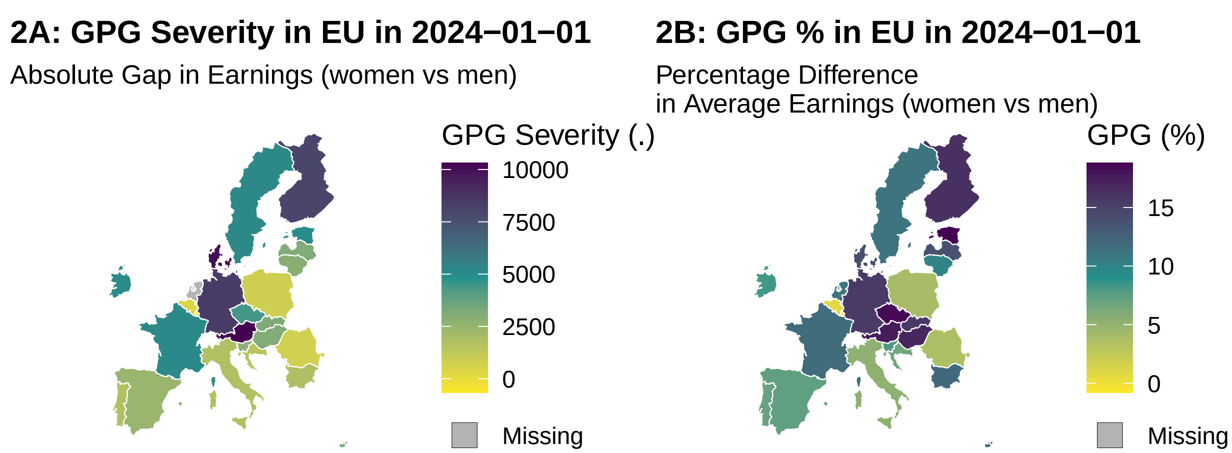
- This plot shows us the average unadjusted Gender Pay Gap as a percentage for the same European regions. When assessing the pay gap in terms of percentages, it makes it easier for us to compare and see the relative inequality across the four European regions. A declining line therefore suggests that the relative difference between women's and men's earnings is narrowing. We can also observe that despite different wage levels, such as between Eastern and Western Europe, these two regions have a similar pay gap relative to the associated incomes in those regions.

Relevance to the social problem

- Together, the two plots shown above clearly separate the relative and the absolute dimensions of the Gender Pay Gap issue. Whilst an European region might show a shrinking percentage gap over time, the monetary severity still remains large, if salaries in that region are high (e.g. Western Europe). The observation of these trends over time will help us properly assess whether policy efforts from the EU, such as equal pay legislation, are effective, and whether economic changes and disruptions (e.g. the 2008 GFC and COVID-19) have affected the Gender Pay Gap in different ways across the EU.

3.4 Visualisation of Spatial Variation

The figure for Visualisation of Spatial Variation consists of two Choropleth maps showcasing all EU countries, both of which are using the most recent year of data available from our merged dataset. The maps' colour gradient ranges from yellow (low values) to green (middle values) to purple (high values). Both maps showcase the latest year and the latest date in our Datasets - 1st of January 2024.



Map 2A: Gender Pay Gap Severity in Euros

- The fill colour in this map shows the Gender Pay Gap Severity in Euros for each EU country. Darker purple indicates a larger absolute gap in earnings for women. The map also directly shows us which countries are the main culprits behind this problem - i.e. which countries' women experience the heaviest financial and economical penalty as a result of the Gender Pay Gap. Even if the percentage gap in those countries appears moderate (e.g. Sweden, which is a high salary economy), the severity is indeed still high in monetary terms.

Map 2B: Gender Pay Gap in %

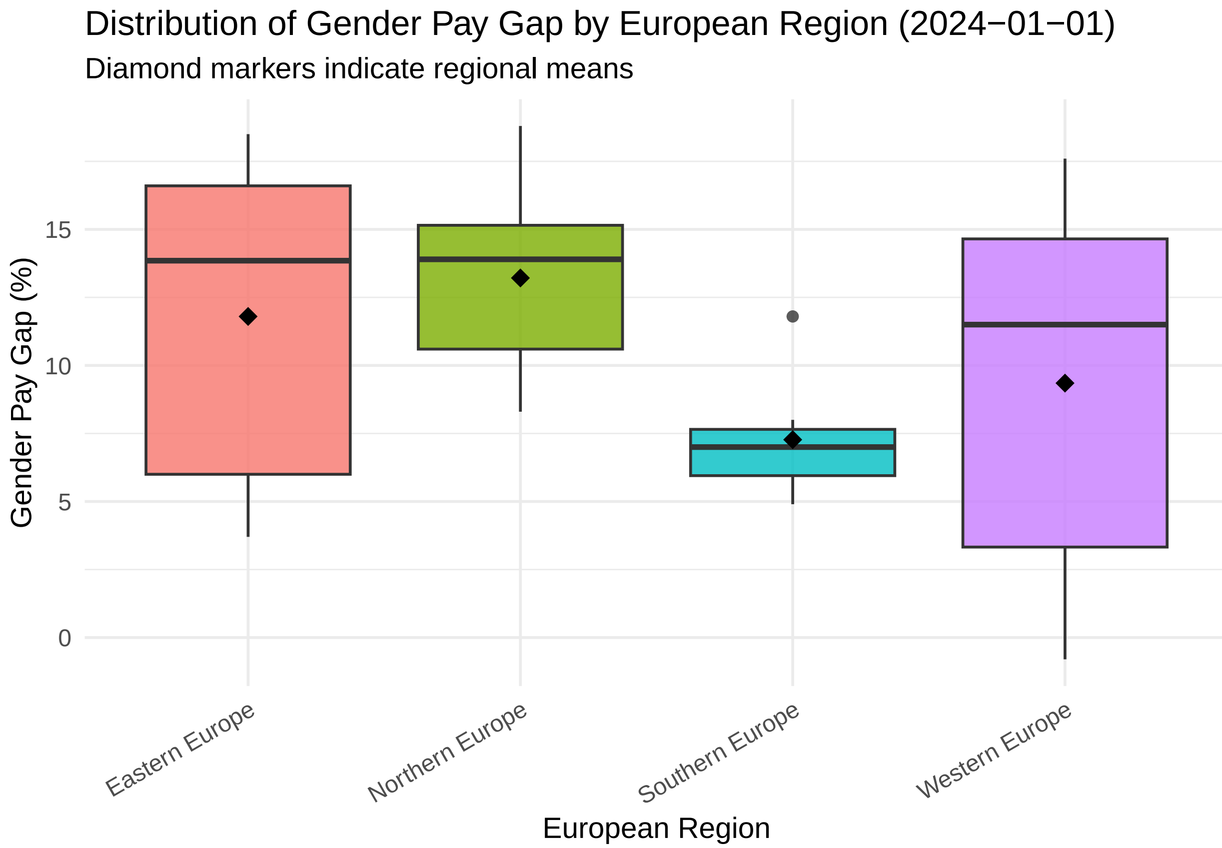
- The fill colour in this map shows the unadjusted Gender Pay Gap as a percentage for each EU country. This map, on the other hand, reflects the relative inequality of how much less women earn compared to men, for each country. This map also helps us to identify EU countries with the most unequal wage structure - regardless of the general wage level. We can also notice that some countries have both a high GPG Severity and a high GPG in %: Germany, Austria, Finland and Denmark are on the top of such list.

Relevance to the social problem

- This Spatial Variation analysis reveals rather striking regional patterns across the EU. The Western European region has both high GPG Severity and high GPG in %, as average salaries are higher in that given region, while conversely, Southern Europe is in general the most equal amongst our examined regions, with having both the lowest GPG Severity and lowest GPG in %.
- By comparing the two maps, we can see that the Gender Pay Gap is indeed not uniform across the EU, and that effective policies must certainly take into consideration the relative gap (GPG in %) experienced and the absolute economic harm (GPG Severity) done to women. Furthermore, our maps enable quick identification of 'GPG Hotspots', which require more urgent attention.

3.5 Visualize sub-population variation

The boxplot illustrates the distribution of the Gender Pay Gap across four European regions in 2024. By presenting both the median and mean values for each region, the figure provides a clearer basis for comparing regional patterns of gender pay inequality across Europe.

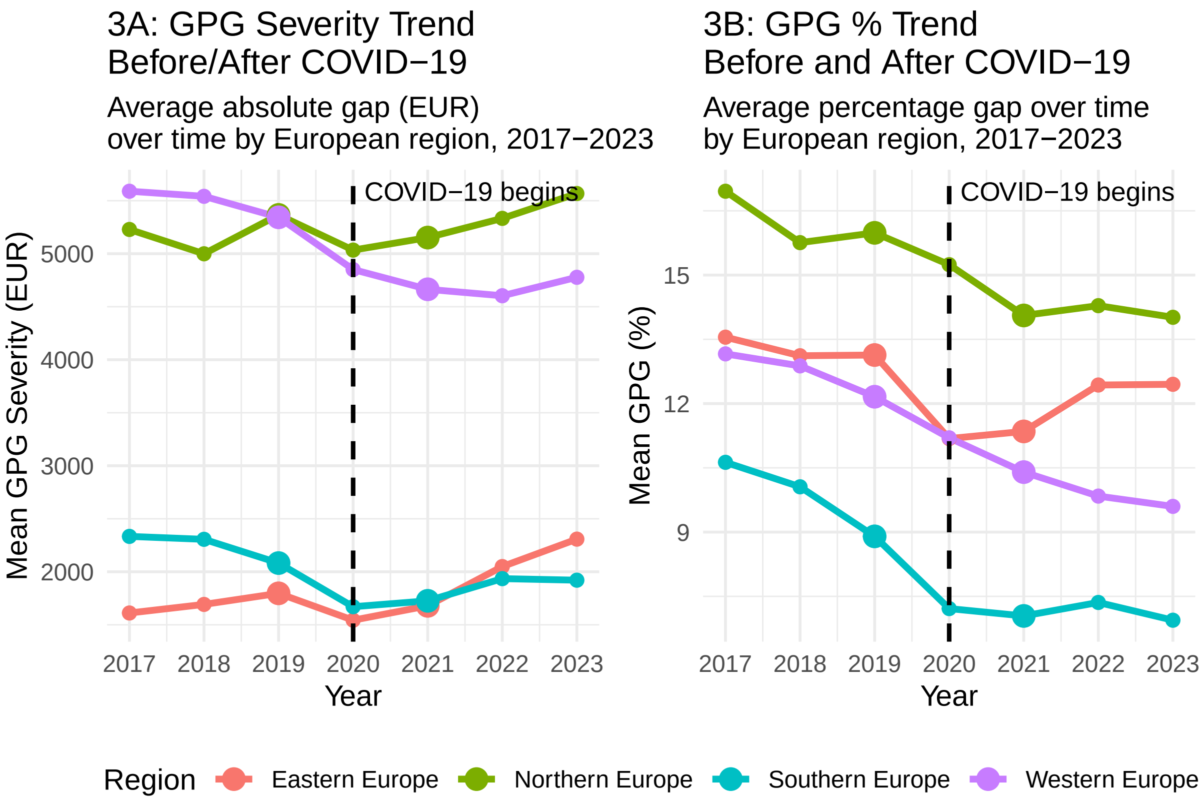


It is clearly shown that Northern Europe has the highest median Gender Pay Gap which clearly shows that most of the countries recording value above 10%. By contrast, Southern Europe has the lowest Gender Pay Gap and it also shows the smallest variation between countries. According to the box plot about Western Europe, it indicates relatively high Gender Pay Gap values, although some lower observations reduce the regional mean. The greatest variation is seen in Western Europe, indicating that gender pay inequality varies greatly among the nations in this region.

In conclusion, these results show that the Gender Pay Gap is not evenly distributed across Europe, which makes the analysis relevant to the social issue of gender pay inequality. Regional differences may reflect variations in labour market structures, economic conditions, and social policies. In addition, the wide range observed in Western Europe suggests that regional averages may not fully represent the experiences of individual countries

3.6 Event analysis

This event analysis examines the Gender Pay Gap before and after the COVID-19 . The dashed vertical line represents the year 2020, when COVID-19 began to significantly affect European labour markets. Comparing the trends before and after this event helps us understand whether the pandemic was associated with changes in gender pay inequality



The line graphs indicate that Southern Europe performs relatively well in general. Throughout the study period, it consistently records the lowest Gender Pay Gap percentages and the lowest GPG Severity values. This suggests that both the relative pay gap and the actual income difference between men and women are smaller in this region.

In comparison, Northern Europe clearly shows the inequality in payment between the woman and the man. Although the Gender Pay Gap percentage decreased slightly since 2020, GPG severity remains high and increased after Covid-19. Although the relative pay gap has narrowed slightly, the actual income gap between men and women remains quite significant

Western Europe shows the most noticeable improvement after COVID-19. Both the Gender Pay Gap percentage and euro-based GPG Severity decline over time. It is indicate that reduction in gender pay inequality in both relative and absolute terms. Eastern Europe presents a more complex pattern. Although the Gender Pay Gap percentage decreases, the actual salary gap continues to widen

Part 4 - Discussion

The main goal of this study is determining whether the gender pay gap actually differs across European regions, and whether things have changed over time. Using Eurostat data, we investigated whether regional differences exist and how they have shifted over time.

As you can see within our research, what came out of it was clear: the gap hasn't gone away anywhere, but it looks very different depending on where you are. Western and Northern Europe tend to have the highest wages overall, yet that's also where the actual euro difference between what men and women take home tends to be biggest. Southern Europe flips that around a bit, smaller gaps and less income lost to inequality, even if average pay is lower across the board.

Over the years covered, the gap has slowly narrowed, which is something at least. But no region has closed it completely, and that says something important: just growing the economy or pushing wages up isn't going to cut it on its own.

4.1 Interpretation of Finding

Even the wealthiest regions haven't escaped big pay gaps in euro terms. Plenty of Western and Northern European countries score well on gender equality indexes. (European Commission, 2024)

Occupational segregation likely explains a lot of this. Across Europe, men and women still cluster in different parts of the job market, men lean toward higher-paying fields like finance, tech, and engineering, while women are more often found in education, healthcare, and social work (European Institute for Gender Equality, 2023). That sorting alone can keep pay gaps alive even where strong equal-pay laws are on the books.

None of this means location itself causes the gap. It's more that geography reflects a mix of institutional, economic, demographic, and cultural forces layered on top of each other. Earlier research points to things like childcare access, parental leave policies, how labour markets are structured, and broader social norms as key drivers of these outcomes for men and women (European Institute for Gender Equality, 2023). So the patterns here should be read as associations, not proof of cause and effect.

4.2 Limitations

A few limitations are worth flagging. First, we relied on Eurostat's unadjusted pay gap figure, which doesn't account for things like education, occupation, work experience, or career breaks. That means we can't really say how much of the gap comes down to discrimination versus differences in who's working where (Eurostat, 2024).

Second, the data is country-level and aggregated, which is great for spotting broad regional patterns but glosses over what's happening within countries, pay gaps can swing widely by industry, company, or demographic group, so national averages only tell part of the story.

Lumping countries into four regions also simplifies things in a way that might hide real institutional and economic differences between, say, Germany and Sweden, or Italy and Greece. Welfare systems, labour policies, and cultural attitudes toward gender roles vary a lot across Europe, and regional buckets can't fully account for that.

Lastly, this is a pattern-spotting study, not a causal one. The regional differences are real, but other unmeasured factors could be shaping both wage levels and the pay gap itself, so the results should be read with that caveat in mind.

4.3 Suggestions for Future Research

Going forward, it'd be worth digging into what's actually driving these regional differences, things like education levels, labour force participation, childcare access, and parental leave setups. Zooming in on individual countries could also shed light on how specific labour markets work and whether certain policies are actually moving the needle.

It'd also help to compare the adjusted and unadjusted pay gap measures side by side, to tease apart how much of the gap can be explained by factors like job type or experience, and how much is left unexplained.

All in all, there's still a lot to figure out here. More research is needed to really get a handle on what's keeping the gender pay gap alive across Europe, and to find approaches that might actually help close it.

Part 5 - Reproducibility

5.1 Github repository link

Public GitHub Repository Link: https://github.com/YessicaR-lab/ProgrammingForEcon

5.2 Reference list

Blau, F. D., & Kahn, L. M. (2017). The gender wage gap: Extent, trends, and explanations. *Journal of Economic Literature, 55*(3), 789–865. https://doi.org/10.1257/jel.20160995

European Commission. (2024a). *Gender equality strategy.* https://commission.europa.eu/strategy-and-policy/policies/justice-and-fundamental-rights/gender-equality/gender-equality-strategy_en

European Commission. (2024b). *2024 Report on Gender Equality in the European Union.* European Commission.

European Institute for Gender Equality. (2023). *Gender Equality Index 2023: Towards a green transition in transport and energy.* EIGE.

Eurostat. (2024). *Gender pay gap statistics.* https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Gender_pay_gap_statistics

Eurostat. (n.d.). *Average full-time adjusted salary per employee (nama_10_fte)* [Data set]. https://ec.europa.eu/eurostat/databrowser/view/nama_10_fte___custom_13597179/bookmark/table?lang=en&bookmarkId=c29eed24-377e-4763-aaf0-0419906d2ecd

Eurostat. (n.d.). *Gender pay gap in unadjusted form (tesem180)* [Data set]. https://ec.europa.eu/eurostat/databrowser/view/tesem180/default/table?lang=en

International Labour Organization. (2024). *Global wage report 2024–25.* https://www.ilo.org/publications